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On time adaptive critical variable exponent vectorial diffusion flows and their applications in image processing I: Analysis

V.B. Surya Prasath[a,b,c], Dmitry Vorotnikov[d,*]

[a] Department of Computer Science, University of Missouri-Columbia, MO 65211, USA
[b] Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center (CCHMC), Cincinnati, OH 45229, USA
[c] Department of Biomedical Informatics, College of Medicine, University of Cincinnati, OH, USA
[d] CMUC, Department of Mathematics, University of Coimbra, 3001-501 Coimbra, Portugal

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ABSTRACT

Variable exponent spaces have found interesting applications in real world problems. In imaging models, the variable exponent can approach the critical value 1 and this poses unique challenges in proving existence of solutions for the corresponding PDEs. In this work, we develop some new functional framework to study time-dependent parabolic variable exponent flows. Specifically, we consider bounded vectorial partial variation ($BVPV$) space and its variable exponent counterpart. We then prove the existence of weak solutions to the critical vectorial $p(t,x)$-Laplacian flow in the variable exponent $BVPV$ space. For time-independent critical vectorial $p(x)$-Laplacian flow we obtain a unique semigroup solution. Our results are in particular valid in the scalar case and solve a long standing open problem.
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1. Introduction

Variable exponent function spaces have been considered by researchers in the past motivated by applications in the modeling of electrorheological fluids [33]. Fundamental results on standard Lebesgue and Sobolev variable exponent spaces have now been well established [18]. Variable-exponent-based regularization turned out to be useful for the restoration of noisy digital images. The simplest variable exponent model for imaging applications is the following one. Assuming we have a noisy version $u_0 : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$ of a latent

* Corresponding author.
E-mail addresses: prasaths@missouri.edu (V.B.S. Prasath), mitvorot@mat.uc.pt (D. Vorotnikov).

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image u the following minimization problem is advocated to find a "restored" image:

$$\min_u \int_\Omega \left(|\nabla u|^{p(x)} + |u - u_0|^2\right) dx. \tag{1}$$

Another useful model is the parabolic counterpart of this problem (gradient flow):

$$\begin{cases} \partial_t u(t,x) = \operatorname{div}[|\nabla u(t,x)|^{p(x)-2}\nabla u(t,x)] + (u_0(x) - u(t,x)), \\ \dfrac{\partial u}{\partial \nu}(t,x) = 0, \ x \in \partial\Omega, \\ u(0,x) = u_0(x). \end{cases} \tag{2}$$

The pioneering works in this direction include [9,16], and we can refer, e.g., to [14,25,15] for some developments. The smoothness requirement depends on the domain of application and typically in image restoration one expects to obtain piecewise constant image with constant regions separated by discontinuities (edges). Here a spatially varying point-wise function $p(x) \in [1,2]$ needs to be designed [31] to automatically switch between the canonical cases of edge preserving total variation [32] ($p = 1$) and isotropically smoothing Tikhonov [36] ($p = 2$) regularizations, which are widely employed in various ill-posed inverse problems from image processing [7].

There exists a large amount of works on the mathematical analysis of the elliptic and parabolic PDEs involving a variable exponent, see e.g. the monographs [5,18] and the references therein. Let us also mention the papers [10–12] where related parabolic problems with non-standard growth are treated. Majority of those works, however, make the assumption that the variable exponent p is bounded away from 1 as the critical value 1 presents difficulties in proving the well-posedness. Typically, the *log-Hölder* regularity of the exponent is also assumed. The functional framework for treating the critical elliptic case was developed in [20]. Consequently, the authors of [21] managed to prove solutions to (1) in the variable exponent BV-Sobolev space. However, the existence of solutions in the critical parabolic case has remained an open problem which we are going to solve in this paper.

If the exponent is critical everywhere, we recover the celebrated total variation flow (with a lower order "fidelity" term). The total variation flow possesses various types of generalized solutions [4,3,30,13]. However, the transition between the reflexive Sobolev ($p > 1$) and BV ($p = 1$) regimes creates technical difficulties which we overcome in this paper.

Since we are motivated by noise removal applications in multichannel (color, multispectral) imagery, we are going to consider a vectorial $p(t,x)$-Laplacian flow, but the results are of course true in the more conventional scalar case.

We are going to study the following time adaptive critical variable exponent flow. Fix a measurable variable exponent $p : (0,T) \times \Omega \to [1, p_+]$. We look for the solutions $u : (0,T) \times \Omega \to \mathbb{R}^N$ to the Neumann problem of the form

$$\begin{cases} \partial_t u(t,x) = \operatorname{div}[|\nabla u(t,x)|^{p(t,x)-2}\nabla u(t,x)] + f(t,x,u(t,x)), \\ \dfrac{\partial u}{\partial \nu}(t,x) = 0, \ x \in \partial\Omega, \\ u(0,x) = u_0(x). \end{cases} \tag{3}$$

We provide a complete functional framework for the critical variable exponent $p(t,x)$ (allowing the exponent to take value 1 on a subset of $(0,T) \times \Omega$); in particular, we define the space of bounded vectorial partial variation (BVPV) and its exponent version $BVPV^{p(\cdot)}$. This allows us to properly define weak solutions to (3) and to prove their existence. The question whether those solutions are unique is open. However, we also obtain a unique semigroup solution for the time-independent vectorial $p(x)$-Laplacian. We point out that our results do not require log-Hölder regularity of the exponent; it should just be slightly better than measurable (namely, to be Y-*semicontinuous*, see the next section for this notion).

In the forthcoming companion paper we provide experimental numerical results which prove efficiency of this model for multichannel imaging.

We organize the rest of the paper as follows. In Section 2, we make some basic conventions and recall some preliminary facts concerning the BV and variable exponent spaces. In Section 3, we introduce and study some new functional spaces which are apt for the critical $p(t,x)$-Laplacian flows. In Section 4, we apply these ideas to show existence of solutions to (3). In Appendix, we prove an auxiliary result on solvability of nonlinear equations in Hilbert triples.

2. Conventions and preliminaries

Throughout the paper, $\Omega \subset \mathbb{R}^n$ is a fixed bounded domain with Lipschitz boundary, $Q = (0,T) \times \Omega$ is the corresponding parabolic cylinder, we also denote $Q_t = (0,t) \times \Omega$ for $0 < t < T$. For a vector function $u : Q \to \mathbb{R}^N$, we write $\nabla u(t,x) \in \mathbb{M}$ for its Jacobi matrix, where $\mathbb{M} := \mathbb{R}^{N\times n}$. For a matrix function $A : Q \to \mathbb{M}$, the divergence operator acts as $(\operatorname{div} A)_i = \sum_{j=1}^n \frac{\partial A_{ij}}{\partial x_j}$. The Frobenius norm of a matrix $A \in \mathbb{M}$ is denoted by $|A| = \sqrt{Tr(A^\top A)}$. The scalar product of vectors from $\mathbb{R}^N$ and the component-wise scalar product in $\mathbb{M}$ is denoted by a dot. The Lebesgue measure of a set is denoted by $|\cdot|$, and $(\cdot,\cdot)$ and $\|\cdot\|$ are the scalar product and Euclidean norm in $L^2(\Omega,\mathbb{R}^N)$ or in $L^2(\Omega,\mathbb{M})$. Since Ω is bounded, when we refer to Radon measures, we always mean finite Radon measures.

Given a Hilbert space H, the *subdifferential* [6,19] of a functional $\Phi : H \to \mathbb{R} \cup \{+\infty\}$ is the multivalued map

$$\partial\Phi : \operatorname{dom}\partial\Phi \subset H \multimap H,$$

$$\partial\Phi(u) = \{w \in H : \Phi(u+h) \geq \Phi(u) + (w,h)_H,\ \forall h \in H\}. \tag{4}$$

The set $\operatorname{dom}\partial\Phi$ consists of those $u \in H$ for which the right-hand side of (4) is non-empty. When Φ is differentiable, $\partial\Phi$ is single-valued and coincides with the usual gradient $\operatorname{grad}_H \Phi : H \to H$.

We give here some basic definitions of the bounded variation and variable exponent spaces, for further details we refer to [18,20,24]. Let $L^0(\Omega)$ be the set of measurable scalar functions on $\Omega \subset \mathbb{R}^n$. Fix a function $p \in L^0(\Omega)$ (the variable exponent). Denote

$$p_- = ess\inf_\Omega p(x), \quad \text{and} \quad p_+ = ess\sup_\Omega p(x).$$

For our needs it suffices to assume that

$$1 \leq p_- \leq p_+ < +\infty.$$

Consider the modular $\rho(u) = \int_\Omega |u(x)|^{p(x)}\,dx$. Then

$$L^{p(\cdot)}(\Omega) := \{u \in L^0(\Omega) | \exists\lambda > 0 :\ \rho(u/\lambda) < +\infty\}.$$

The Luxemburg–Nakano norm on $L^{p(\cdot)}$ is $\|u\|_{L^{p(\cdot)}} := \inf\{\lambda > 0 : \rho(u/\lambda) \leq 1\}$. We recall that

$$L^{p_+}(\Omega) \subset L^{p(\cdot)}(\Omega) \subset L^{p_-}(\Omega), \tag{5}$$

and the embeddings are continuous.

The Orlicz–Sobolev space is

$$W^{1,p(\cdot)}(\Omega) := \{u \in L^{p(\cdot)}(\Omega) : \nabla u \in L^{p(\cdot)}(\Omega)^n\}$$

with $\|u\|_{W^{1,p(\cdot)}} := \|u\|_{L^{p(\cdot)}} + \||\nabla u|\|_{L^{p(\cdot)}}$. Then

$$W^{1,p_+}(\Omega) \subset W^{1,p(\cdot)}(\Omega) \subset W^{1,p_-}(\Omega), \tag{6}$$

and the embeddings are continuous.

The bounded variation space is defined as follows: a function $u \in BV(\Omega)$ if $u \in L^1(\Omega)$, and if its total variation

$$TV(u) := \sup\left\{\int_\Omega u \operatorname{div}\phi : \phi \in C_0^1(\Omega)^n, \|\phi\|_{L^\infty(\Omega)^n} \le 1\right\}$$

is finite; the norm in $BV(\Omega)$ is given by

$$\|u\|_{BV} := \|u\|_{L^1} + TV(u).$$

The distributional gradient ∇u of $u \in BV(\Omega)$ is a vector-valued Radon measure, and hence the total variation $|\nabla u|$ (in the measure-theoretic sense) of this measure is a Radon measure. It is called the total variation measure and is sometimes also denoted by $TV(u)$. One can check that $TV(u)(\Omega) = TV(u)$.

In the case when $p_- = 1$, the Orlicz–Sobolev space defined above is not reflexive, and it is plausible to use the space $BV^{p(\cdot)}(\Omega)$ defined below in various applications.

Definition 1. A measurable function $p : \Omega \to [1, +\infty)$ is Y-*semicontinuous* if the following property holds: given a converging sequence $\{x_k\} \subset \Omega$,

$$\lim_{k\to\infty} x_k \in \Omega,\ \lim_{k\to\infty} p(x_k) = 1 \Longrightarrow p\left(\lim_{k\to\infty} x_k\right) = 1.$$

Assume that our variable exponent $p(x)$ is a Y-semicontinuous function. Let

$$Y := \{x \in \Omega : p(x) = 1\}$$

be the critical set where the exponent takes the value 1. It is obviously closed in the relative topology of Ω.

Remark 2. It is usually assumed that the exponent $p(\cdot)$ is at least lower semicontinuous [21] in the definition of $BV^{p(\cdot)}(\Omega)$. However, our weaker assumption of Y-semicontinuity is enough. Our usage of the letter Y to denote the 1-locus of the variable exponent (and in the notion of Y-semicontinuity, accordingly) goes back to the Finnish school [20,21]. "Yksi" is Finnish for "one".

We now put $BV^{p(\cdot)}(\Omega) := BV(\Omega) \cap W^{1,p(\cdot)}(\Omega \setminus Y)$, and consider the nonnegative number

$$TV^{p(\cdot)}(u) := TV(u)(Y) + \int_{\Omega\setminus Y} |\nabla u|^{p(x)}\, dx.$$

Then the norm in $BV^{p(\cdot)}$ is

$$\|u\|_{BV^{p(\cdot)}(\Omega)} := \|u\|_{L^{p(\cdot)}} + \inf\left\{\lambda > 0 : TV^{p(\cdot)}(u/\lambda) \le 1\right\}.$$

All the spaces discussed above are Banach ones.

Remark 3. If $u \in BV(\Omega)^N$ is a vector function, then we can set

$$TV(u) := \sup\left\{\int_\Omega u \cdot \operatorname{div}\Phi : \Phi \in C_0^1(\Omega;\mathbb{M}), \|\Phi\|_{L^\infty(\Omega;\mathbb{M})} \le 1\right\}. \tag{7}$$

Moreover (see Section 3 for more general considerations), the components of the distributional Jacobi matrix ∇u are signed Radon measures on Ω, and the total variation (Radon) measure is determined by the formula

$$TV(u)(E) = \sup\left\{\sum_k |\nabla u(E_k)|\right\}, \tag{8}$$

where the supremum is taken over all Borel partitions $\{E_k\}$ of a Borel set $E \subset \Omega$.

3. The functional framework: Sobolev-$BVPV$ spaces

The standard variable exponent function spaces (which were recalled in the previous section) are suitable for stationary problems. We now develop some additional functional framework in order to study parabolic variable exponent flows.

Let $\mathcal{M}$ be the continuous dual of the space $C_0(Q)^N$ of continuous vector functions on Q that vanish on the boundary of the cylinder Q. By the Riesz duality [24, Theorem B114], it can be also viewed as the Banach space of N-vectorial signed Radon measures on Q equipped with the total variation norm

$$\|v\|_{\mathcal{M}} = |v|(Q), \tag{9}$$

where the Radon measure $|v|$ is the total variation of the measure v. We recall that

$$|v|(E) = \sup\left\{\sum_k |v(E_k)|\right\}, \tag{10}$$

where the supremum is taken over all Borel partitions $\{E_k\}$ of a Borel set $E \subset Q$. For $v \in \mathcal{M}$, we define the *vectorial partial variation* of v as

$$VPV(v) = \sup_{\Phi \in C_0^1(Q;\mathbb{M}):\, |\Phi| \le 1} \langle v, \operatorname{div} \Phi\rangle_{\mathcal{M}\times C_0(Q)^N}. \tag{11}$$

Observe that if $v \in L^1(0,T;W_1^1(\Omega))^N$, then $VPV(v) = \int_0^T \int_\Omega |\nabla v(t,x)|\,dx\,dt$.

Remark 4. We dub (11) "partial" variation because it accounts for the partial derivatives in space and not in time. Perhaps this is not the best wording, but we adhere to it since we already opted for it in [30] in a similar situation.

Define the "bounded vectorial partial variation space" $BVPV$ as

$$\left\{v \in \mathcal{M} \middle| \|v\|_{BVPV} := \|v\|_{\mathcal{M}} + VPV(v) < +\infty\right\}.$$

The following proposition is straightforward by the Riesz duality:

Proposition 5. *For any $v \in BVPV$, the components of its distributional Jacobi matrix ∇v are signed Radon measures on Q.*

For any $v \in BVPV$ and a Borel set $E \subset Q$, consider the nonnegative scalar

$$VPV(v)(E) = \sup\left\{\sum_k |\nabla v(E_k)|\right\}, \tag{12}$$

where the supremum is taken over all Borel partitions $\{E_k\}$ of E. Then [24, Proposition B75 and Theorem B114] and Proposition 5 imply

Proposition 6. *For any $v \in BVPV$, $VPV(v)$ is a Radon measure on Q, and $VPV(v)(Q) = VPV(v)$.*

Remark 7. It is quite customary and rather convenient in the theory of functions of bounded variation to denote the total variation of a function and the corresponding Radon measure by the same symbol, cf. Section 2. We opt for a similar abuse of notation in (12) and below in (22).

For every open set $U \subset Q$ and $v \in BVPV$, applying [24, Theorem B114] to the measure $\nabla v \llcorner U \in (C_0(U;\mathbb{M}))^*$, we infer that

$$VPV(v)(U) = \sup_{\Phi \in C_0^\infty(U;\mathbb{M}):\, |\Phi| \le 1} \langle v, \operatorname{div} \Phi \rangle_{\mathcal{M}(U)\times C_0(U)^N}. \tag{13}$$

Owing to lower semicontinuity of suprema, we can derive from (13) the following fact:

Proposition 8. *For any weakly-* converging (in $\mathcal{M}$) sequence $\{v_m\} \subset BVPV$, one has*

$$VPV(v) \le \liminf_{m\to+\infty} VPV(v_m) \tag{14}$$

where $v = \lim v_m$. Moreover,

$$VPV(v)(U) \le \liminf_{m\to+\infty} VPV(v_m)(U) \tag{15}$$

for every open set $U \subset Q$.

Then (14) yields

Proposition 9. *$BVPV$ is a Banach space.*

We now fix a Y-semicontinuous scalar function $p : Q \to [p_-, p_+] \subset [1, +\infty)$ (the variable exponent). Note that the critical set

$$Y := [p(t,x) = 1] \tag{16}$$

is closed in the relative topology of Q.

The evolutionary Orlicz–Sobolev space $W^{0,1}_{p(\cdot)}(Q \setminus Y)^N$ is defined as

$$W^{0,1}_{p(\cdot)}(Q \setminus Y)^N := \{v \in L^{p(\cdot)}(Q \setminus Y)^N : \nabla v \in L^{p(\cdot)}(Q \setminus Y;\mathbb{M})\}.$$

It is easy to establish

Proposition 10. *The space $W^{0,1}_{p(\cdot)}(Q \setminus Y)^N$ equipped with the norm*

$$\|v\|_{W^{0,1}_{p(\cdot)}(Q\setminus Y)^N} := \||v|\|_{L^{p(\cdot)}(Q\setminus Y)} + \||\nabla v|\|_{L^{p(\cdot)}(Q\setminus Y)}$$

is a Banach space. It is reflexive provided $p_- > 1$ and $p_+ < +\infty$.

We define the subset $BVPV^{p(\cdot)}$ of $\mathcal{M}$ as follows: a vectorial Radon measure v belongs to $BVPV^{p(\cdot)}$ provided $VPV(v) < \infty$, the restriction $v \llcorner (Q\setminus Y)$ is absolutely continuous with respect to the Lebesgue measure, and the Radon–Nikodym derivative

$$v|_{Q\setminus Y} := \frac{d(v \llcorner (Q\setminus Y))}{dx \otimes dt}$$

belongs to $W^{0,1}_{p(\cdot)}(Q \setminus Y)^N$. This definition is expressed by the following descriptive but sloppy formula:

$$BVPV^{p(\cdot)} = BVPV \cap W^{0,1}_{p(\cdot)}(Q \setminus Y)^N. \tag{17}$$

Given a function $v \in BVPV^{p(\cdot)}$, we define the nonnegative number

$$VPV^{p(\cdot)}(v) = VPV(v)(Y) + \int_{Q\setminus Y} |\nabla v(t,x)|^{p(t,x)}\, dx\, dt. \tag{18}$$

Proposition 11. *$BVPV^{p(\cdot)}$ is a Banach space, being equipped with the norm*

$$\|v\|_{BVPV_{p(\cdot)}} := |v|(Y) + \|v\|_{L^{p(\cdot)}(Q\setminus Y)^N} + \inf\left\{\lambda > 0 : VPV^{p(\cdot)}(v/\lambda) \le 1\right\}, \tag{19}$$

or with the equivalent norm

$$\|v\|_{BVPV^{p(\cdot)}} := \|v\|_{BVPV} + \|v\|_{W^{0,1}_{p(\cdot)}(Q\setminus Y)^N}. \tag{20}$$

Proof. It is clear that (20) is a norm on $BVPV^{p(\cdot)}$. Let $\{v_k\}$ be a Cauchy sequence in this norm. By Proposition 9, it admits a limit v in $BVPV$. Hence, $v_k \llcorner (Q\setminus Y) \to v \llcorner (Q\setminus Y)$ in total variation. By Proposition 10, $v_k|_{Q\setminus Y}$ has a limit in $W^{0,1}_{p(\cdot)}(Q\setminus Y)^N$, which should coincide with $v|_{Q\setminus Y}$. Thus, $BVPV^{p(\cdot)}$ is a Banach space.

Observe now that $VPV^{p(\cdot)}(v) < +\infty$ for $v \in BVPV^{p(\cdot)}$. Hence,

$$\inf\left\{\lambda > 0 : VPV^{p(\cdot)}(v/\lambda)\right\} < +\infty$$

for all $v \in BVPV^{p(\cdot)}$. Furthermore, $\inf\left\{\lambda > 0 : VPV^{p(\cdot)}(v/\lambda)\right\}$ is a seminorm on $BVPV^{p(\cdot)}$ by an argument similar to the proof of [18, Theorem 2.1.7]. Consequently, (19) is a norm on $BVPV^{p(\cdot)}$.

It remains to conclude that both norms are equivalent since, due to (5),

$$\begin{aligned}
&\|v\|_{BVPV_{p(\cdot)}} \to 0 \\
&\qquad \iff |v_k|(Y) + VPV(v_k)(Y) + \int_{Q\setminus Y} |v_k(t,x)|^{p(t,x)} + |\nabla v_k(t,x)|^{p(t,x)}\, dx\, dt \to 0 \\
&\qquad \iff |v_k|(Q) + VPV(v_k) + \int_{Q\setminus Y} |v_k(t,x)|^{p(t,x)} + |\nabla v_k(t,x)|^{p(t,x)}\, dx\, dt \to 0 \\
&\qquad \iff \|v\|_{BVPV^{p(\cdot)}} \to 0. \quad \square
\end{aligned} \tag{21}$$

Remark 12. The space $BVPV^{p(\cdot)}$ is relevant for image processing applications since it allows for jumps on the critical set Y even if this critical set is merely a null set in Q (e.g., a hypersurface in Q).

For any function $v \in BVPV^{p(\cdot)}$, we define the measure $VPV^{p(\cdot)}(v)$ on Q by letting

$$VPV^{p(\cdot)}(v) := VPV(v) \llcorner Y + |\nabla v(t,x)|^{p(t,x)}\, dx\, dt \llcorner (Q\setminus Y). \tag{22}$$

The following observation is clear in view of (18) and Remark 7:

Proposition 13. *For any $v \in BVPV^{p(\cdot)}$, $VPV^{p(\cdot)}(v)$ is a Radon measure on Q, and $VPV(v)^{p(\cdot)}(Q) = VPV^{p(\cdot)}(v)$.*

The following lower-semicontinuity property of $VPV^{p(\cdot)}$ is crucial in the applications.

Theorem 14. *For any open set $U \subset Q$, and for any weakly-* converging (in $\mathcal{M}$) sequence $\{v_m\} \subset BVPV^{p(\cdot)}$, $v = \lim v_m$, one has*

$$VPV^{p(\cdot)}(v)(U) \le \liminf_{m\to+\infty} VPV^{p(\cdot)}(v_m)(U). \tag{23}$$

In particular, if $U = Q$ and the limit inferior is finite, then $v \in BVPV^{p(\cdot)}$.

Proof. Assume that $U = Q$. Mutatis mutandis, the same argument works for $U \neq Q$.

Let the limit inferior in (23) be finite (otherwise the theorem holds trivially). By Proposition 8, $v \in BVPV$.

For any $\epsilon > 0$, consider the sets

$$Y_\epsilon := \left\{ s_0 \in Q, \inf_{s \in \partial Q \cup Y} |s_0 - s| \leq \epsilon \right\}, \tag{24}$$

which are closed in the relative topology of Q. Let $U_\epsilon := \overset{\circ}{Y_\epsilon}$.

Then

$$|\partial Y_\epsilon| = |\partial U_\epsilon| = 0 \tag{25}$$

(at least up to a countable set of ϵ's). Due to Y-semicontinuity of p,

$$\inf_{(t,x) \notin Y_\epsilon} p(t,x) > p_-. \tag{26}$$

Moreover, $\cap_{\epsilon>0}(Y_\epsilon \setminus Y) = \varnothing$, whence

$$|Y_\epsilon \setminus Y| \to 0, \tag{27}$$

$$\int_{U_\epsilon \setminus Y} |\nabla v(t,x)|^{p(t,x)} \, dx \, dt \to 0 \tag{28}$$

as $\epsilon \to 0$.

The spaces $W^{0,1}_{p(\cdot)}(Q \setminus Y_\epsilon)^N$ are reflexive by Proposition 10. Consequently, passing to a subsequence if necessary,

$$v_m|_{Q \setminus Y_\epsilon} \to v|_{Q \setminus Y_\epsilon} \text{ weakly in } W^{0,1}_{p(\cdot)}(Q \setminus Y_\epsilon)^N. \tag{29}$$

Hence, by [18, Theorem 2.2.8],

$$\begin{aligned} \int_{Q \setminus Y_\epsilon} |\nabla v(t,x)|^{p(t,x)} \, dx \, dt &\leq \liminf_{m \to +\infty} \int_{Q \setminus Y_\epsilon} |\nabla v_m(t,x)|^{p(t,x)} \, dx \, dt \\ &\leq \liminf_{m \to +\infty} VPV^{p(\cdot)}(v_m). \end{aligned} \tag{30}$$

Letting $\epsilon \to 0$, and remembering (27), we infer

$$\int_{Q \setminus Y} |\nabla v(t,x)|^{p(t,x)} \, dx \, dt \leq \liminf_{m \to +\infty} \int_{Q \setminus Y} |\nabla v_m(t,x)|^{p(t,x)} \, dx \, dt. \tag{31}$$

Hence, $v \in BVPV^{p(\cdot)}$.

By Proposition 8,

$$\begin{aligned} VPV(v)(U_\epsilon) &\leq \liminf_{m \to +\infty} VPV(v_m)(U_\epsilon) \\ &= \liminf_{m \to +\infty} \left[VPV(v_m)(Y) + \int_{U_\epsilon \setminus Y} |\nabla v_m(t,x)| \, dx \, dt \right] \\ &\leq \liminf_{m \to +\infty} \left[VPV(v_m)(Y) + \int_{U_\epsilon \setminus Y} \left(1 + |\nabla v_m(t,x)|^{p(t,x)} \right) \, dx \, dt \right]. \end{aligned} \tag{32}$$

Employing (25), (26), (30), (32), and superadditivity of $\liminf$, we conclude that

$$VPV^{p(\cdot)}(v) = VPV(v)(Y) + \int_{Q \setminus Y} |\nabla v(t,x)|^{p(t,x)} \, dx \, dt$$

$$
\begin{aligned}
&\le VPV(v)(U_\epsilon) + \int_{Q\setminus Y_\epsilon} |\nabla v(t,x)|^{p(t,x)}\,dx\,dt + \int_{U_\epsilon\setminus Y} |\nabla v(t,x)|^{p(t,x)}\,dx\,dt \\
&\le \liminf_{m\to+\infty}\left[VPV(v_m)(Y) + |U_\epsilon\setminus Y| + \int_{U_\epsilon\setminus Y} |\nabla v_m(t,x)|^{p(t,x)}\,dx\,dt\right] \\
&\quad + \liminf_{m\to+\infty}\int_{Q\setminus Y_\epsilon} |\nabla v_m(t,x)|^{p(t,x)}\,dx\,dt + \int_{U_\epsilon\setminus Y} |\nabla v(t,x)|^{p(t,x)}\,dx\,dt \\
&\le |U_\epsilon\setminus Y| + \int_{U_\epsilon\setminus Y} |\nabla v(t,x)|^{p(t,x)}\,dx\,dt + \liminf_{m\to+\infty} VPV^{p(\cdot)}(v_m),
\end{aligned}
\tag{33}
$$

and (23) follows due to (27) and (28). □

4. The critical vectorial $p(t,x)$-Laplacian flow

4.1. Weak solutions

Fix a Y-semicontinuous variable exponent $p : Q \to [1, p_+]$ with $Y = [p(t,x) = 1]$ being the critical set in Q. We are interested in defining and studying the solutions $(u, Z) : Q \to \mathbb{R}^N \times \mathbb{M}$ to the following Neumann problem

$$
\begin{cases}
\partial_t u(t,x) = \operatorname{div} Z(t,x) + f(t,x,u(t,x)), \\
Z(t,x) = |\nabla u(t,x)|^{p(t,x)-2}\nabla u(t,x), \\
Z(t,x)\nu(x) = 0,\ \ x \in \partial\Omega, \\
u(0,x) = u_0(x).
\end{cases}
\tag{34}
$$

The fidelity term is a measurable function $f : Q \times \mathbb{R}^N \to \mathbb{R}^N$ of at most linear growth in the sense

$$
|f(t,x,u)| \le \tilde{f}(t,x) + C|u| \tag{35}
$$

for some measurable function $\tilde{f} : Q \to \mathbb{R}$. The new unknown Z is introduced for technical convenience.

Definition 15. Let $u_0 \in L^2(\Omega)^N$, $\tilde{f} \in L^2(Q)$. A pair of functions (u, Z) from the class

$$
\begin{aligned}
&u \in C_w([0,T]; L^2(\Omega)^N) \cap BVPV^{p(\cdot)} \cap W^1_{(p_+)'}(0,T;[L^2 \cap W^{1,p_+}(\Omega)^N]^*), \\
&Z \in L^{(p_+)'}(Q;\mathbb{M}) \cap L^\infty(Y;\mathbb{M}),\ \ \|Z\|_{L^\infty(Y;\mathbb{M})} \le 1,
\end{aligned}
\tag{36}
$$

is called a weak solution to problem (34) if
(i) for all $v \in L^2 \cap W^{1,p_+}(\Omega)^N$,

$$
\left\langle \frac{du}{dt}, v \right\rangle + \big(Z, \nabla v\big) = (f(\cdot,u), v) \tag{37}
$$

a.e. on $(0,T)$;
(ii) for all $w \in L^{p_+}(0,T;W^{1,p_+}(\Omega))^N \cap W^{1,1}(0,T;L^2(\Omega))^N$ and any open set U, $Y \subset U \subset Q$, one has

$$
\begin{aligned}
&\|u(T) - w(T)\|^2 + 2\int_Q \partial_t w(t,x)\cdot u(t,x)\,dx\,dt + 2\,VPV^{p(\cdot)}(u)(U) \\
&\quad \le \|u_0 - w(0)\|^2 + \|w(T)\|^2 - \|w(0)\|^2 + 2\int_U Z(t,x)\cdot\nabla w(t,x)\,dx\,dt \\
&\qquad + 2\int_{Q\setminus U} |\nabla w(t,x)|^{p(t,x)-2}\nabla w(t,x)\cdot(\nabla w(t,x) - \nabla u(t,x))\,dx\,dt \\
&\qquad + 2\int_Q f(t,x,u(t,x))\cdot(u(t,x) - w(t,x))\,dx\,dt
\end{aligned}
\tag{38}
$$

(iii) the initial condition

$$u(0) = u_0 \tag{39}$$

holds in the space $L^2(\Omega)^N$, i.e., $u(t) \to u_0$ weakly in $L^2(\Omega)^N$ as $t \to 0_+$.

Observation 16. Let us present a motivation for this definition of solution to (34). Further motivation is discussed in Observation 25.

Consider a pair (u, Z) of sufficiently regular functions satisfying (36), (37), (38), (39). Then we can rewrite (38) in the form

$$\begin{aligned} 2\int_Q \partial_t u(t,x)\cdot(u(t,x)-w(t,x))\,dx\,dt + 2\int_U |\nabla u(t,x)|^{p(t,x)}\,dx\,dt \\ \le 2\int_U Z(t,x)\cdot\nabla w(t,x)\,dx\,dt \\ + 2\int_{Q\setminus U} |\nabla w(t,x)|^{p(t,x)-2}\nabla w(t,x)\cdot(\nabla w(t,x)-\nabla u(t,x))\,dx\,dt \\ + 2\int_Q f(t,x,u(t,x))\cdot(u(t,x)-w(t,x))\,dx\,dt. \end{aligned} \tag{40}$$

Due to (37) it reduces to

$$\begin{aligned} \int_U |\nabla u(t,x)|^{p(t,x)}\,dx\,dt \le \int_Q Z(t,x)\cdot(\nabla u(t,x)-\nabla w(t,x))\,dx\,dt \\ + \int_U Z(t,x)\cdot\nabla w(t,x)\,dx\,dt \\ + \int_{Q\setminus U} |\nabla w(t,x)|^{p(t,x)-2}\nabla w(t,x)\cdot(\nabla w(t,x)-\nabla u(t,x))\,dx\,dt. \end{aligned} \tag{41}$$

Testing (41) by $w \equiv u$, we get

$$\int_Y |\nabla u(t,x)|\,dx\,dt \le \int_U |\nabla u(t,x)|^{p(t,x)}\,dx\,dt \tag{42}$$

$$\le \int_U Z(t,x)\cdot\nabla u(t,x)\,dx\,dt, \tag{43}$$

Letting $U = U_\epsilon$ where U_ϵ was defined during the proof of Theorem 14, and allowing $\epsilon \to 0$, we derive

$$\int_Y |\nabla u(t,x)|\,dx\,dt \le \int_Y Z(t,x)\cdot\nabla u(t,x)\,dx\,dt, \tag{44}$$

On the other hand,

$$|\nabla u| \ge Z\cdot\nabla u \tag{45}$$

a.e. in Y, whence

$$|\nabla u| = Z\cdot\nabla u. \tag{46}$$

Since $|Z| \le 1$ a.e. in Y, (46) yields

$$Z = \frac{\nabla u}{|\nabla u|} = |\nabla u|^{p(\cdot)-2}\nabla u \tag{47}$$

a.e. in Y. Rigorously speaking, if $|Y \cap [\nabla u = 0]| > 0$, then (47) merely holds in the following subdifferential sense

$$Z \in \partial\psi(\nabla u) \tag{48}$$

a.e. in Y, where $\partial\psi$ is the subdifferential (cf. Section 2) of the function $\psi : \mathbb{M} \to \mathbb{R}$, $\psi(A) = |A|$.

Substituting expression (46) to the left-hand side of (41), we derive that

$$\begin{aligned}\int_{Q\setminus U} &(Z - |\nabla w(t,x)|^{p(t,x)-2}\nabla w(t,x)) \cdot (\nabla u(t,x) - \nabla w(t,x))\, dx\, dt \\ &\geq \int_{U\setminus Y} |\nabla u(t,x)|^{p(t,x)}\, dx\, dt - \int_{U\setminus Y} Z(t,x)\cdot\nabla u(t,x)\, dx\, dt.\end{aligned} \tag{49}$$

Letting $U = U_\epsilon$, $\epsilon \to 0$, we get

$$\int_{Q\setminus Y} (Z - |\nabla w(t,x)|^{p(t,x)-2}\nabla w(t,x)) \cdot (\nabla u(t,x) - \nabla w(t,x))\, dx\, dt \geq 0. \tag{50}$$

By a Minty argument, this yields

$$Z = |\nabla u|^{p(\cdot)-2}\nabla u \tag{51}$$

a.e. in $Q \setminus Y$.

Integrating be parts in (37), we infer

$$(\partial_t u - \operatorname{div} Z - f(\cdot,u), v) + \int_{\partial\Omega} v \cdot Z\nu\, d\mathcal{H}^{n-1} = 0. \tag{52}$$

Testing (52) by any smooth v compactly supported in Ω, we deduce the first equation in (34). Consequently, both integrals in (52) are identically zero. By arbitrariness of v, $Z\nu = 0$ $\mathcal{H}^{n-1}$-a.e. in $\partial\Omega$.

The main result of the paper is

Theorem 17. *If $u_0 \in L^2(\Omega)^N$, $\tilde{f} \in L^2(Q)$, and the function $v \mapsto v \cdot f(t,x,v)$ is concave for a.a. $(t,x) \in Q$. Then there exists a weak solution to (34).*

In order to prove Theorem 17 we first need to study the following auxiliary problem:

$$\begin{cases} \partial_t u + \delta\mathcal{B}u = \operatorname{div}\left(\dfrac{|\nabla u|^{2p(\cdot)-2}\nabla u}{\epsilon + |\nabla u|^{p(\cdot)}}\right) + f(\cdot,u), \\ u(0,x) = u_0(x). \end{cases} \tag{53}$$

Here ϵ, δ are positive parameters; the operator $\mathcal{B}$ and the boundary conditions are to be specified below.

Consider the Hilbert triple (see Appendix for definitions) $(H^r(\Omega)^N, L^2(\Omega)^N, ((H^r(\Omega))^*)^N)$ where $r > \left(1 + \frac{n}{2}\right)(p_+ + 1)$ is a fixed number. Denote by $\mathcal{B}$ the Riesz bijection between the spaces $(H^r)^N$ and $((H^r)^*)^N$. The weak form of (53) (with a certain implicit boundary condition which is of no importance to us) is the following Cauchy problem, where the first equality is understood as an ODE in the space $((H^r)^*)^N$, whereas the second equality is in the sense of the space $(L^2)^N$:

$$u' + \delta\mathcal{B}u = \mathcal{Q}(t,u) + f(t,\cdot,u), \quad u|_{t=0} = u_0. \tag{54}$$

Here the operator $\mathcal{Q} : (0,T) \times C^1(\overline{\Omega})^N \to ((H^r(\Omega))^*)^N$ is determined by the duality

$$\langle \mathcal{Q}(t,u), w\rangle = -\left(\frac{|\nabla u|^{2p(t,\cdot)-2}\nabla u}{\epsilon + |\nabla u|^{p(t,\cdot)}}, \nabla w\right), \quad \forall w \in (H^r)^N,$$

and the prime stands for the time derivative.

Lemma 18. *Let $u_0 \in L^2(\Omega)^N$, $\tilde{f} \in L^2(Q)$. The Cauchy problem (54) admits a solution u in the class*

$$L^2(0,T;(H^r(\Omega))^N) \cap H^1(0,T;((H^r(\Omega))^*)^N) \cap C([0,T];(L^2(\Omega))^N). \tag{55}$$

The solution satisfies the following inequality:

$$\begin{aligned}
&\|u(t_*) - w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x)\cdot u(t,x)\,dx\,dt + 2\,VPV^{p(\cdot)}(u)(Q_{t_*}\cap U)\\
&\quad\le \|u_0 - w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*}\cap U} \frac{|\nabla u(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}}\nabla u(t,x)\cdot\nabla w(t,x)\,dx\,dt\\
&\qquad + 2\int_{Q_{t_*}\setminus U} \frac{|\nabla w(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla w(t,x)|^{p(t,x)}}\nabla w(t,x)\cdot(\nabla w(t,x) - \nabla u(t,x))\,dx\,dt\\
&\qquad + 2\int_{Q_{t_*}} f(t,x,u(t,x))\cdot(u(t,x) - w(t,x))\,dx\,dt\\
&\qquad + 2\epsilon|Q_{t_*}\cap U| + \frac{\delta}{2}\|w\|^2_{L^2(0,t_*;(H^r(\Omega))^N)}
\end{aligned} \tag{56}$$

for every $t_* \in [0,T]$, every open set U, $Y \subset U \subset Q$, and for every sufficiently regular test function $w : \overline{Q} \to \mathbb{R}^N$.

Proof. 1. We are going to apply Lemma 26 with

$$X = H^r(\Omega)^N,\ Y = L^2(\Omega)^N,\ V = C^1(\overline{\Omega})^N,\ \mathcal{Q}_M = \mathcal{Q},\ \mathcal{Q}_L = f,$$

$$\mathcal{A} = \delta\mathcal{B},\ m = p_+,\ \theta = \frac{1}{p_+ + 1}.$$

A classical interpolation inequality in Sobolev Hilbert scale [8] gives

$$\|u\|_{H^{r\theta}} \le C\,\|u\|^{\theta}_{H^r}\,\|u\|^{1-\theta}_{L^2}. \tag{57}$$

Since $r\theta > 1 + \frac{n}{2}$, we have by Sobolev embedding that

$$\|u\|_{C^1(\overline{\Omega})} \le C\|u\|_{H^{r\theta}}, \tag{58}$$

so (104) is satisfied. Moreover, the embedding $H^r(\Omega)^N \subset C^1(\overline{\Omega})^N$ is compact. The operators $\mathcal{Q}, f : (0,T)\times C^1(\overline{\Omega})^N \to ((H^r(\Omega))^*)^N$ satisfy the Carathéodory condition. Moreover, f satisfies condition (105) since

$$\|f(t,\cdot,u)\|_{((H^r(\Omega))^*)^N} \le \|f(t,\cdot,u)\|_{L^2(\Omega)^N} \le \|\tilde{f}(t)\|_{L^2(\Omega)} + C\|u\|_{C^1(\overline{\Omega})^N}, \tag{59}$$

and $\tilde{f} \in L^2(0,T;L^2(\Omega))$. Furthermore, $\mathcal{Q}$ satisfies (106) since

$$\begin{aligned}
\|\mathcal{Q}(t,u)\|_{((H^r(\Omega))^*)^N} &\le \|\mathcal{Q}(t,u)\|_{\left((H^1(\Omega))^*\right)^N} \le \left\|\frac{|\nabla u|^{2p(\cdot)-2}\nabla u}{\epsilon + |\nabla u|^{p(\cdot)}}\right\|_{L^2(\Omega)^N}\\
&\le \left\|1 + |\nabla u|^{p_+-1}\right\|_{L^2(\Omega)} \le C(1 + \|u\|^{p_+}_{C^1(\overline{\Omega})^N}).
\end{aligned} \tag{60}$$

Thus, the conditions of Lemma 26 are met, and the existence of solution u follows immediately.

2. We now fix a sufficiently smooth function $w : \overline{Q} \to \mathbb{R}$ and an open set U, $Y \subset U \subset Q$. Test (54) with $2(u-w)(t)$, $0 \le t \le t_* \le T$, in the sense of $(((H^r(\Omega))^*)^N, H^r(\Omega)^N)$-duality, and integrate in time to obtain

$$\|u(t_*) - w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x)\cdot u(t,x)\,dx\,dt + 2\int_{Q_{t_*}\cap U} \frac{|\nabla u(t,x)|^{2p(t,x)}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}}\,dx\,dt$$

$$\leq \|u_0 - w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*}\cap U} \frac{|\nabla u(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}} \nabla u(t,x)\cdot\nabla w(t,x)\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}\setminus U} \frac{|\nabla u(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}} \nabla u(t,x)\cdot(\nabla w(t,x) - \nabla u(t,x))\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}} f(t,x,u(t,x))\cdot(u(t,x) - w(t,x))\,dx\,dt - 2\delta\int_0^{t_*} \langle \mathcal{B}u(t), u(t) - w(t)\rangle\,dt. \tag{61}$$

Note that the following finite-dimensional inequality holds

$$\left(\frac{|\xi|^{2p-2}}{\epsilon + |\xi|^p}\xi - \frac{|\eta|^{2p-2}}{\epsilon + |\eta|^p}\eta\right)\cdot(\xi - \eta) \geq 0, \quad p \geq 1, \epsilon > 0,\ \xi, \eta \in \mathbb{M}. \tag{62}$$

This follows from the convexity of the potential

$$\xi \mapsto \frac{|\xi|^p}{p} - \frac{\epsilon}{p}\log(\epsilon + |\xi|^p), \quad \xi \in \mathbb{M}.$$

Applying Cauchy's inequality, inequality (62) and making some rearrangements, we derive from (61) that

$$\|u(t_*) - w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x)\cdot u(t,x)\,dx\,dt + 2\int_{Q_{t_*}\cap U} |\nabla u(t,x)|^{p(t,x)}\,dx\,dt$$
$$\leq \|u_0 - w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*}\cap U} \frac{|\nabla u(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}} \nabla u(t,x)\cdot\nabla w(t,x)\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}\setminus U} \frac{|\nabla w(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla w(t,x)|^{p(t,x)}} \nabla w(t,x)\cdot(\nabla w(t,x) - \nabla u(t,x))\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}} f(t,x,u(t,x))\cdot(u(t,x) - w(t,x))\,dx\,dt$$
$$+\ 2\epsilon\int_{Q_{t_*}\cap U} \frac{|\nabla u(t,x)|^{p(t,x)}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}}\,dx\,dt + \frac{\delta}{2}\int_0^{t_*} \langle \mathcal{B}w(t), w(t)\rangle\,dt, \tag{63}$$

and (56) follows. □

Observation 19. In Lemma 18, the solution u satisfies the a priori bound

$$\|u\|_{L^2(0,T;(H^r(\Omega))^N)} + \|u\|_{H^1(0,T;((H^r(\Omega))^*)^N)} + \|u\|_{C([0,T];(L^2(\Omega))^N)} + \|u\|_{L^{2p_+ + 2}(0,T;(C^1(\overline{\Omega}))^N)} \leq C, \tag{64}$$

which comes from the proof of Lemma 26. We claim that the constant C here does not depend on ϵ (but may depend on δ). This follows from the observation that the operator $\mathcal{Q}_M = \mathcal{Q}$ satisfies estimate (106) uniformly in ϵ, see (60), and that $\mathcal{Q}_L = f$ is independent of ϵ, δ. Moreover, from (113) we can derive that

$$\delta\,\|u\|^2_{L^2(0,T;(H^r(\Omega))^N)} \leq C, \tag{65}$$

where C is independent of ϵ, δ.

Proof of Theorem 17. 1. Let u be a solution to (54) in the sense of Lemma 18. Set

$$Z = \frac{|\nabla u(t,x)|^{2p(t,x)-2}}{\epsilon + |\nabla u(t,x)|^{p(t,x)}}\nabla u(t,x).$$

Then

$$\|Z\|_{L^\infty(Y;\mathbb{M})} \leq 1, \tag{66}$$

and

$$\|Z\|^{(p_+)'}_{L^{(p_+)'}(Q;\mathbb{M})} \le \int_Q |\nabla u(t,x)|^{(p_+)'(p(t,x)-1)}\,dx\,dt$$
$$\le \int_Q (1+|\nabla u(t,x)|^{p'(t,x)(p(t,x)-1)})\,dx\,dt = |Q| + VPV^{p(\cdot)}(u) \le C. \tag{67}$$

The last constant does not depend on ϵ (but may depend on δ) due to (64).

2. Since (u,Z) is a solution to (54) in the sense of Lemma 18, for all sufficiently smooth functions $v,w:\overline{Q}\to\mathbb{R}^N$ and for every open set U, $Y\subset U\subset Q$, we have

$$\left\langle \frac{du}{dt}, v\right\rangle + \delta\langle \mathcal{B}u, v\rangle + \big(Z,\nabla v\big) = (f(\cdot,u),v) \tag{68}$$

a.e. on $(0,T)$,

$$\|u(t_*)-w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x)\cdot u(t,x)\,dx\,dt + 2\,VPV^{p(\cdot)}(u)(Q_{t_*}\cap U)$$
$$\le \|u_0-w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*}\cap U} Z(t,x)\cdot\nabla w(t,x)\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}\setminus U} \frac{|\nabla w(t,x)|^{2p(t,x)-2}}{\epsilon+|\nabla w(t,x)|^{p(t,x)}}\nabla w(t,x)\cdot(\nabla w(t,x)-\nabla u(t,x))\,dx\,dt$$
$$+\ 2\int_{Q_{t_*}} f(t,x,u(t,x))\cdot(u(t,x)-w(t,x))\,dx\,dt$$
$$+\ 2\epsilon|Q_{t_*}\cap U| + \frac{\delta}{2}\|w\|^2_{L^2(0,t_*;(H^r(\Omega))^N)} \tag{69}$$

for all $t_*\in[0,T]$, and

$$u(0)=u_0. \tag{70}$$

To prove the theorem, we are going to pass to the limit in (68), (69), (70), first as $\epsilon\to 0$, and then as $\delta\to 0$.

3. Take a sequence of solutions (u_k,Z_k) to (54) with $\epsilon=\epsilon_k\to 0$. Due to (64), (67), without loss of generality we have

$$u_k\to u \text{ weakly in } L^2(0,T;(H^r(\Omega))^N), \tag{71}$$
$$u_k\to u \text{ weakly}* \text{ in } L^\infty(0,T;(L^2(\Omega))^N), \tag{72}$$
$$u_k'\to u' \text{ weakly in } L^2(0,T;((H^r(\Omega))^*)^N), \tag{73}$$
$$Z_k\to Z \text{ weakly in } L^{(p_+)'}(Q;\mathbb{M}). \tag{74}$$

By the Aubin–Lions–Simon lemma [34],

$$u_k\to u \text{ strongly in } L^2(Q)^N \text{ and in } C([0,T];((H^r(\Omega))^*)^N). \tag{75}$$

Hence, by a classical continuity property of Nemytskii operators [23],

$$f(\cdot,u_k)\to f(\cdot,u) \text{ strongly in } L^2(Q)^N. \tag{76}$$

Moreover, by a result by Simon [34, Corollary 6],

$$u_k\to u \text{ strongly in } L^{2p_+}(0,T;C^1(\overline{\Omega})^N). \tag{77}$$

Consequently, employing (5),

$$\nabla u_k \to \nabla u \text{ strongly in } L^{p(\cdot)}(Q;\mathbb{M}), \tag{78}$$

and

$$VPV^{p(\cdot)}(u_k)(V) \to VPV^{p(\cdot)}(u)(V) \tag{79}$$

for any Borel set $V \subset Q$. We can thus pass to the limit as $k \to \infty$ in the third term in (69). We can also pass to the limit inferior, keeping the sign of the inequality, in the first member of (69), since this term is lower semicontinuous, see the δ-limit below in the proof for the details of a similar reasoning. By the Lebesgue dominated convergence theorem,

$$\frac{|\nabla w|^{2p(\cdot)-2}}{\epsilon_k + |\nabla w|^{p(\cdot)}} \nabla w \to |\nabla w|^{p(\cdot)-2} \nabla w$$

in $L^2(Q \setminus U)^N$. Hence, passing to the limit in (69) we get

$$\begin{aligned}
&\|u(t_*) - w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x) \cdot u(t,x)\, dx\, dt + 2\, VPV^{p(\cdot)}(u)(Q_{t_*} \cap U) \\
&\quad \le \|u_0 - w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*} \cap U} Z(t,x) \cdot \nabla w(t,x)\, dx\, dt \\
&\qquad + 2\int_{Q_{t_*} \setminus U} |\nabla w(t,x)|^{p(t,x)-2} \nabla w(t,x) \cdot (\nabla w(t,x) - \nabla u(t,x))\, dx\, dt \\
&\qquad + 2\int_{Q_{t_*}} f(t,x,u(t,x)) \cdot (u(t,x) - w(t,x))\, dx\, dt + \frac{\delta}{2} \|w\|^2_{L^2(0,t_*;(H^r(\Omega))^N)}
\end{aligned} \tag{80}$$

for all $t_* \in [0,T]$. Note that (65) and (66) still hold for the limit.

4. Taking $U = Q$, $w \equiv 0$ in (80), and remembering (65), we derive

$$\|u(t_*)\|^2 + 2\, VPV^{p(\cdot)}(u)(Q_{t_*}) \le C(1+\delta) + \int_0^{t_*} \|u(t)\|^2\, dt \tag{81}$$

for all $t_* \in [0,T]$, whence by Grönwall's lemma

$$\|u(t_*)\|^2 + VPV^{p(\cdot)}(u) \le C, \tag{82}$$

where C does not depend on δ (we assume that the possible range of δ is bounded). Then, by (67) and (79),

$$\|Z\|_{L^{(p_+)'}(Q;\mathbb{M})} \le C. \tag{83}$$

Moreover, (82) implies

$$\int_Q |u(t,x)| + |\nabla u(t,x)|\, dx\, dt \le C + \int_Q |\nabla u(t,x)|^{p(t,x)}\, dx\, dt \le C. \tag{84}$$

Observe also that since $H^r \subset W^{1,p_+}$, relations (68), (65), (82), (35) yield

$$\|u'\|_{L^{\min((p_+)',2)}(0,T;((H^r(\Omega))^*)^N)} \le C. \tag{85}$$

All the constants C here are δ-independent.

5. As a result of the previous steps, we can find a sequence of pairs (u_m, Z_m) satisfying (68), (70), (80) with $\delta = \delta_m \to 0$. Due to the δ-independent estimates (82), (83), (85), without loss of generality we may assume that

$$u_m \to u \text{ weakly}* \text{ in } L^\infty(0,T;(L^2(\Omega))^N), \tag{86}$$

$$u_m' \to u' \text{ weakly in } L^{\min((p_+)',2)}(0,T;((H^r(\Omega))^*)^N), \tag{87}$$

$$Z_m \to Z \text{ weakly in } L^{(p_+)'}(Q;\mathbb{M}). \tag{88}$$

By the Aubin–Lions–Simon lemma [34] and bounds (84), (85),

$$u_m \to u \text{ strongly in } L^1(Q)^N, \tag{89}$$

and thus without loss of generality

$$u_m \to u \text{ a.e. in } Q. \tag{90}$$

As above, we can pass to the limit in (68), (70) and in the linear and constant members of (80). Note that the second member in (68) goes to zero by virtue of (65), and the last member of (80) also goes to zero. For a.a. $t_* \in (0,T)$, we can pass to the limit inferior, keeping the sign of the inequality, in the first member of (80) (we test (80) by smooth compactly supported nonnegative scalar functions of time, and pass to the limit in the sense of scalar distributions, see [26,37,30] for technical details in similar situations). By Theorem 14, $u \in BVPV^{p(\cdot)}$ and we can pass to the limit inferior in the $VPV^{p(\cdot)}$-term.

It remains to pass to the limit in the penultimate term of (80) involving the fidelity. By [29, Theorem 2.6], (86) and (90) yield

$$f(\cdot, u_m) \to f(\cdot, u) \text{ weakly in } L^2(Q)^N. \tag{91}$$

Consider the functional

$$F : L^2(Q)^N \to \mathbb{R},\ F(u) = \int_{Q_{t_*}} f(t,x,u(t,x)) \cdot u(t,x)\,dx\,dt.$$

A classical result on Nemytskii operators [23,29] implies that F is bounded and continuous. Since the function $v \mapsto v \cdot f(t,x,v)$ is concave, F is concave and therefore weakly upper semicontinuous. This fact together with (86) and (91) gives the opportunity to pass to the limit superior in the penultimate term of (80), maintaining the sign of the inequality.

Hence, in the limit we get

$$\begin{aligned}
&\|u(t_*) - w(t_*)\|^2 + 2\int_{Q_{t_*}} \partial_t w(t,x) \cdot u(t,x)\,dx\,dt + 2\,VPV^{p(\cdot)}(u)(Q_{t_*} \cap U) \\
&\quad \le \|u_0 - w(0)\|^2 + \|w(t_*)\|^2 - \|w(0)\|^2 + 2\int_{Q_{t_*}\cap U} Z(t,x) \cdot \nabla w(t,x)\,dx\,dt \\
&\qquad + 2\int_{Q_{t_*}\setminus U} |\nabla w(t,x)|^{p(t,x)-2} \nabla w(t,x) \cdot (\nabla w(t,x) - \nabla u(t,x))\,dx\,dt \\
&\qquad + 2\int_{Q_{t_*}} f(t,x,u(t,x)) \cdot (u(t,x) - w(t,x))\,dx\,dt
\end{aligned} \tag{92}$$

for a.a. $t_* \in (0,T)$. By a Lions–Magenes lemma [38, Lemma 2.2.6],

$$u \in C_w([0,T];L^2(\Omega)^N).$$

Therefore, the first term of (92) is lower semicontinuous in time. Since $VPV^{p(\cdot)}(u)$ is a Radon measure on Q, the third term is also lower semicontinuous in time. The remaining members are continuous in time. Hence, (92) holds for all $t_* \in [0,T]$, in particular, for $t_* = T$. Thus, (u, Z) satisfies (37), (38), (39). By density, the test functions v and w in (37) and (38) can be taken from the spaces indicated in Definition 15. □

4.2. Semigroup solutions

In this section we assume that p does not depend on time, i.e. we have a Y-semicontinuous variable exponent $p : \Omega \to [1, p_+]$, and $Y = [p(x) = 1]$ is the critical set in Ω. To fix the ideas and avoid unnecessary techicalities, we also assume that $f \equiv 0$. We are thus left with the following Neumann problem:

$$\begin{cases} \partial_t u(t,x) = \operatorname{div} Z(t,x), \\ Z(t,x) = |\nabla u(t,x)|^{p(x)-2} \nabla u(t,x), \\ Z(t,x)\nu(x) = 0, \ x \in \partial\Omega, \\ u(0,x) = u_0(x). \end{cases} \tag{93}$$

Consider the functional $\Psi : L^2(\Omega)^N \to \mathbb{R}$,

$$\Psi(u) = \begin{cases} TV(u)(Y) + \displaystyle\int_{\Omega\setminus Y} \frac{|\nabla u|^{p(x)}}{p(x)}\, dx, \ u \in (BV^{p(\cdot)}(\Omega))^N \cap L^2(\Omega)^N, \\ +\infty \qquad\qquad\qquad\qquad\qquad\qquad \text{otherwise.} \end{cases} \tag{94}$$

Getting rid of the auxiliary variable Z, it is easy to observe that (93) is a formal gradient flow of Ψ in $L^2(\Omega)^N$:

$$u' + \operatorname{grad}_{L^2(\Omega)^N} \Psi(u) = 0.$$

However, the functional Ψ is not regular enough to employ this fact immediately. But we still have

Lemma 20. *The functional Ψ is proper, convex, lower-semicontinuous and has a dense domain.*

Proof. The lower-semicontinuity can be checked in a way similar to Theorem 14 (see also [21]). The rest is obvious. □

Lemma 20 implies that the subdifferential $\partial\Psi$ has a dense domain $\operatorname{dom}\partial\Psi$, see [6, Subsection 17.2.2]. For convex lower semicontinuous functionals, the notion of subdifferential is the correct extension of the notion of gradient, see again [6, Subsection 17.2.2]. We can thus give the following definition, which is also motivated by [4,3] and by [1,2].

Definition 21. A function

$$W^{1,\infty}_{loc}((0,+\infty); L^2(\Omega)^N) \cap C([0,+\infty); L^2(\Omega)^N)$$

is called a semigroup solution to (93) provided $u(t) \in \operatorname{dom}\partial\Psi$ for all $t > 0$, $0 \in u' + \partial\Psi(u)$. For a.a. $t > 0$, and $u(0) = u_0$.

By [6, Theorem 17.2.3], we shortly have,

Theorem 22. *For any $u_0 \in L^2(\Omega)^N$, there is a unique semigroup solution to (93).*

Remark 23. A reader might ask why we have spent so much effort in proving existence of a weak solution if the existence of a semigroup one is almost immediate. The first reason is that (in the case when the critical set Y is a nonempty proper subset of Ω) there is no satisfactory characterization of the subdifferential $\partial\Psi$ (for instance, in terms of variational inequalities, cf. [4] where the case $p \equiv 1$ is considered); hence the link between the semigroup solutions and the original problem (93) is not completely clear. The second reason is that the semigroup approach is hardly generalizable to the case of time-dependent exponents p and fidelities f because in this situation the domain of Ψ depends on t, and the infrequent existing treatments of Hilbertian

gradient flows with a time-dependent driving potential (e.g., [22]) are not applicable. One could instead try to apply the well-developed theory of nonlinear semigroups in Banach spaces, which admits time-dependent generators, see e.g. [17,28], but there would be still some technical obstructions: in particular, the domain of the generating operators, no matter how exactly we define them, would vary with time.

Remark 24. Semigroup solutions automatically have plenty of nice properties provided by the gradient flow structure, such as continuous dependence on u_0 and long-time convergence to the infimum of Ψ, we refer to [6, Section 17] for the details.

We finish this section by establishing a formal link between the weak and semigroup solutions.

Observation 25. Let (u, Z) be a sufficiently regular weak solution to (93) in the sense of Definition 15. We claim that u is a (unique) semigroup solution. Indeed, fix an element $h \in (BV^{p(\cdot)}(\Omega))^N \cap L^2(\Omega)^N$. For any $s \geq 1$, let us introduce the function

$$\psi_s : \mathbb{M} \to \mathbb{R},\ \psi_s(A) = \frac{|A|^s}{s}.$$

Then we can rewrite (48) and (51) in the subdifferential form

$$Z(t,x) \in \partial\psi_{p(x)}(\nabla u(t,x)) \tag{95}$$

a.e. in Q. In particular, for a.a. $x \in \Omega \setminus Y$ we have

$$\psi_{p(x)}(\nabla(u+h)(t,x)) \geq \psi_{p(x)}(\nabla u(t,x)) + \nabla h(x) \cdot Z(t,x), \tag{96}$$

whence

$$\int_{\Omega\setminus Y} \frac{|\nabla(u+h)|^{p(x)}}{p(x)}\,dx \geq \int_{\Omega\setminus Y} \frac{|\nabla u|^{p(x)}}{p(x)}\,dx + \int_{\Omega\setminus Y} \nabla h(x) \cdot Z(t,x)\,dx. \tag{97}$$

By [4, Corollary C.7] we have

$$TV(u+h)(Y) \geq \frac{\int_Y Z(t,x)\,d\,\nabla(u+h)}{\|Z\|_{L^\infty(U;\mathbb{M})}} \tag{98}$$

for any open set $U \subset \Omega$ containing Y. Since $Z \cdot \nabla u = |\nabla u|$ a.e. in Y and $\|Z\|_{L^\infty(Y;\mathbb{M})} \leq 1$, provided Z is continuous we infer

$$\begin{aligned} TV(u+h)(Y) &\geq \int_Y Z(t,x)\,d\,\nabla h(x) + \int_Y \nabla u(t,x) \cdot Z(t,x)\,dx \\ &= \int_Y Z(t,x)\,d\,\nabla h(x) + \int_Y |\nabla u(t,x)|\,dx. \end{aligned} \tag{99}$$

Adding (97) and (99), remembering that $\partial_t u(t,x) = \operatorname{div} Z(t,x)$ and $Z\nu = 0$ $\mathcal{H}^{n-1}$-a.e. in $\partial\Omega$, and using [4, Theorem C.9], we deduce

$$\Psi(u+h) \geq \Psi(u) - \int_\Omega \operatorname{div} Z(t,x) \cdot h(x)\,dx = \Psi(u) - \int_\Omega \partial_t u(t,x) \cdot h(x)\,dx. \tag{100}$$

Trivially, (100) also holds for $h \in L^2(\Omega)^N \setminus (BV^{p(\cdot)}(\Omega))^N$. Hence,

$$-u' \in \partial\Psi(u), \tag{101}$$

and u is a semigroup solution.

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Appendix. On solvability of some nonlinear Cauchy problems in Hilbert triples

Assume that there are two Hilbert spaces, $X \subset Y$, with continuous embedding operator $i : X \to Y$, and $i(X)$ is dense in Y. The adjoint operator $i^* : Y^* \to X^*$ is continuous and, since $i(X)$ is dense in Y, one-to-one. Since i is one-to-one, $i^*(Y^*)$ is dense in X^*, and one may identify Y^* with a dense subspace of X^*. Due to the Riesz representation theorem, one may also identify Y with Y^*. We arrive at the chain of inclusions:

$$X \subset Y \equiv Y^* \subset X^*. \tag{102}$$

Both embeddings here are dense and continuous. Observe that in this situation, for $f \in Y, u \in X$, their scalar product in Y coincides with the value of the functional f from X^* on the element $u \in X$:

$$(f, u)_Y = \langle f, u \rangle. \tag{103}$$

Such triples (X, Y, X^*) are called *Hilbert triples* (sometimes also referred to as Gelfand or Lions triples), see, e.g., [35,38] for more details.

Lemma 26. *Let*

$$X \subset Y \subset X^*$$

be a Hilbert triple. Let $\mathcal{A} : X \to X^*$ be a linear continuous operator such that

$$\langle \mathcal{A}u, u \rangle \geq \alpha \|u\|_X^2$$

for all $u \in X$ and some common $\alpha > 0$. Let V be a Banach space such that

$$X \subset V \subset Y$$

where the first embedding is compact and the second is continuous. Assume that both X and V are separable. Assume moreover that

$$\|u\|_V \leq \|u\|_X^{\theta} \|u\|_Y^{1-\theta} \tag{104}$$

for some $\theta \in (0, 1)$. Let

$$\mathcal{Q}_L, \mathcal{Q}_M : (0, T) \times V \to X^*$$

be Carathéodory functions, i.e., $\mathcal{Q}_L(t, \cdot), \mathcal{Q}_M(t, \cdot)$ are continuous for a.a. $t \in (0, T)$ and $\mathcal{Q}_L(\cdot, u), \mathcal{Q}_M(\cdot, u)$ are measurable for all $u \in V$. Assume that

$$\|\mathcal{Q}_L(t, u)\|_{X^*} \leq C(F(t) + \|u\|_V), \tag{105}$$

$$\|\mathcal{Q}_M(t, u)\|_{X^*} \leq C(F(t) + \|u\|_V^m), \ \langle \mathcal{Q}_M(t, u), u \rangle \leq 0, \tag{106}$$

for all $u \in X$ and a.a. $t \in (0,T)$, where $F \in L^2(0,T)$ is independent of u, and $m \geq 1$, $m\theta < 1$. Then the Cauchy problem

$$u'(t) + \mathcal{A}u(t) = \mathcal{Q}_L(t,u(t)) + \mathcal{Q}_M(t,u(t)), \quad u|_{t=0} = u_0, \tag{107}$$

has a solution in the class

$$L^2(0,T;X) \cap H^1(0,T;X^*) \cap L^{2/\theta}(0,T;V) \cap C([0,T];Y) \tag{108}$$

for every $u_0 \in Y$.

Proof. 1. Consider the Banach space

$$W := L^2(0,T;X) \cap H^1(0,T;X^*)$$

with the natural intersection norm. By the Aubin–Lions–Simon lemma [34], the embedding $W \subset L^2(0,T;V)$ is compact. Moreover, by [38, Lemma 2.2.7 and Corollary 2.2.3] any function from W may be considered as an element of $C([0,T];Y)$, and the embedding $W \subset C([0,T];Y)$ is continuous. Due to (104),

$$\|u\|_{L^{2/\theta}(0;T;V)} \leq \|u\|^{\theta}_{L^2(0,T;X)} \|u\|^{1-\theta}_{L^\infty(0,T;Y)}, \tag{109}$$

therefore the embedding $W \subset L^{2/\theta}(0;T;V)$ is also continuous. By [34, Corollary 6], the embedding $W \subset L^{2m}(0;T;V)$ is compact (as $2m < 2/\theta$). The Nemytskii operators

$$Q_L(\cdot,u) : L^2(0,T;V) \to L^2(0,T;X^*),$$

$$Q_M(\cdot,u) : L^{2m}(0,T;V) \to L^2(0,T;X^*)$$

are bounded and continuous by the Lucchetti–Patrone theorem [27,29]. Hence, the Nemytskii operators

$$Q_L(\cdot,u), Q_M(\cdot,u) : W \to L^2(0,T;X^*)$$

are continuous and compact.

2. Consider the family of Cauchy problems

$$u' + \mathcal{A}u = \lambda(\mathcal{Q}_L(t,u) + \mathcal{Q}_M(t,u)), \quad u|_{t=0} = u_0, \tag{110}$$

where $\lambda \in [0,1]$ is a parameter. We are going to prove that the solutions of (110) are a priori bounded in the space W, uniformly in λ. Taking the scalar product in Y of (110) and u a.e. in $(0,T)$, we infer

$$\frac{1}{2}\frac{d}{dt}\|u\|^2_Y + \alpha\|u\|^2_X \leq \lambda\langle \mathcal{Q}_L(t,u) + Q_M(t,u), u\rangle \leq C(F(t) + \|u\|_V)\|u\|_X. \tag{111}$$

Therefore,

$$\frac{d}{dt}\|u\|^2_Y + \alpha\|u\|^2_X \leq C(F^2(t) + \|u\|^2_V) \leq C(F^2(t) + \|u\|^{2\theta}_X \|u\|^{2(1-\theta)}_Y), \tag{112}$$

and by Young's inequality

$$\frac{d}{dt}\|u\|^2_Y + \frac{\alpha}{2}\|u\|^2_X \leq C(F^2(t) + \|u\|^2_Y). \tag{113}$$

By Grönwall's lemma,

$$\|u\|_{L^2(0,T;X)} + \|u\|_{L^\infty(0,T;Y)} \leq C, \tag{114}$$

hence, by (109),

$$\|u\|_{L^{2m}(0,T;V)} \leq C. \tag{115}$$

Consequently, (110) together with (105), (106) yields

$$\|u'\|_{L^2(0,T;X^*)} \leq C. \tag{116}$$

3. By [38, Lemma 3.1.3] and Banach's continuous inverse theorem, the linear operator

$$\tilde{\mathcal{A}} : W \to Y \times L^2(0,T;X^*), \quad \tilde{\mathcal{A}}(u) = (u|_{t=0}, u' + \mathcal{A}u)$$

is continuously invertible. Now (110) can be rewritten as

$$u = \lambda \tilde{\mathcal{A}}^{-1}(u_0, \mathcal{Q}_L(\cdot, u) + \mathcal{Q}_M(\cdot, u)). \tag{117}$$

The operator $\tilde{\mathcal{Q}}(u) := \tilde{\mathcal{A}}^{-1}(u_0, \mathcal{Q}_L(\cdot, u) + \mathcal{Q}_M(\cdot, u))$ is continuous and compact in W, and the solutions of (117) are uniformly bounded in W. By Schaefer's fixed point theorem [19], $\tilde{\mathcal{Q}}$ has a fixed point, which is a solution of (107). □

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